



ICPC SOUTH PACIFIC PRELIMINARY CONTEST LEVEL A

SEPTEMBER 14, 2025

Contest Problems

A: Arachnid Area
B: Broken Heart
C: Corporate Family
D: Dictionary Shuffle
E: Euclidean Extension
F: Firefighting Robots
G: Get Their Attention
H: Humanoid Robot Tango
I : I Get Extra Votes
J : Jumping Between Stations
K: Kiwi Invasion

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This contest contains eleven problems. Good luck.

For problems that state “Your answer should have an absolute or relative error of less than 10^{-9} ”, your answer, x , will be compared to the correct answer, y . If $|x - y| < 10^{-9}$ or $\frac{|x - y|}{|y|} < 10^{-9}$, then your answer will be considered correct.

Definition 1

For problems that ask for a result modulo m :

If the correct answer to the problem is the integer b , then you should display the unique value a such that:

- $0 \leq a < m$
and
 - $(a - b)$ is a multiple of m .
-

Definition 2

A string $s_1 s_2 \dots s_n$ is lexicographically smaller than $t_1 t_2 \dots t_\ell$ if

- there exists $k \leq \min(n, \ell)$ such that $s_i = t_i$ for all $1 \leq i < k$ and $s_k < t_k$
or
 - $s_i = t_i$ for all $1 \leq i \leq \min(n, \ell)$ and $n < \ell$.
-

Definition 3

- Uppercase letters are the uppercase English letters ($A, B, \dots, Z$).
 - Lowercase letters are the lowercase English letters ($a, b, \dots, z$).
-

Definition 4

Unless otherwise specified, the distance between two points (x_0, y_0) and (x_1, y_1) is defined as its Euclidean distance:

$$\sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2}.$$

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Problem A

Arachnid Area

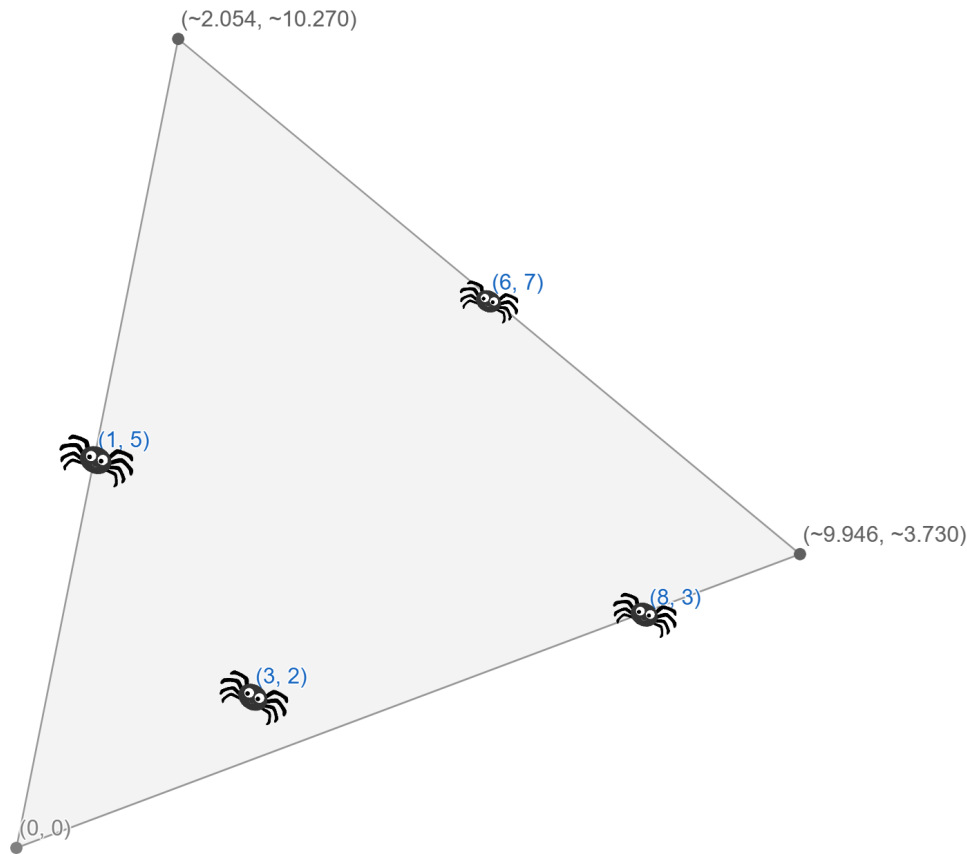
Time limit: 1 second

Lantau Island, Hong Kong, is home to some beautiful nature walks and incredible wildlife. On one of your hikes up the mountain, you stumble upon a fascinating sight, a cluster of giant spiders resting on an invisible web!

After some careful watching, you suspect that the web has a triangular shape anchored at one point, the start of a tree branch at coordinates $(0, 0)$. The spiders themselves seem to be suspended at various points in the air, all lying within the web.

What is the minimum area of the triangular web, anchored at $(0, 0)$, that encloses all of the spiders' positions?

For example, if there are four spiders located at $(1, 5)$, $(6, 7)$, $(3, 2)$, and $(8, 3)$, then the minimum-area triangle that contains all spiders is:



Input

The first line contains an integer n ($2 \leq n \leq 100$), which is the number of spiders.

The next n lines describe the locations of the spiders. Each of these lines contains two integers X and Y ($1 \leq X, Y \leq 1\,000$), indicating that this spider is located at (X, Y) . No two spiders are at the same location.

It is guaranteed that no single line passes through both $(0, 0)$ and every spider.

Output

Display the area of the smallest triangle anchored at $(0, 0)$ that encloses all of the spiders' positions, in square units.

Your answer should have an absolute or relative error of less than 10^{-6} .

Sample Input 1

```
2
1 2
2 1
```

Sample Output 1

```
1.500000000000
```

Sample Input 2

```
4
1 5
6 7
3 2
8 3
```

Sample Output 2

```
47.243243243243
```

Sample Input 3

```
4
1 1
1 4
3 2
5 1
```

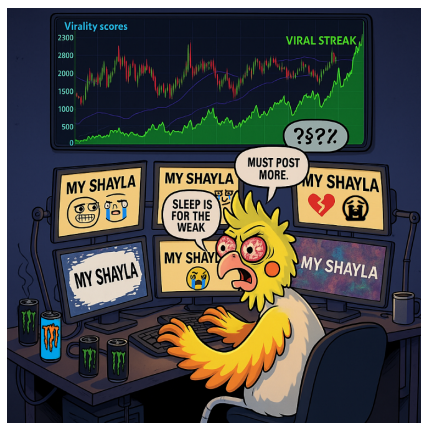
Sample Output 3

```
9.500000000000
```

Problem B

Broken Heart

Time limit: 8 seconds



Our extra epic and absolutely amazing friend (not a paid review) Tom, woke up one day and decided that she must become a part of the canon now! After deep reflection (and three energy drinks), she realized the best way to go viral was to post nonstop “MY SHAYLA 🥺🥺” memes on TicTac, GramGram, YouTuba, AOW, and every other platform in existence.

After many consecutive days of tireless meme posting, Tom collected detailed data on the virality of each of her posts. Each day is assigned a “virality score” measuring how well her content performed. Tom wants to know her longest viral streak. She defines a viral streak as a subarray of days whose arithmetic mean virality (meme mean) is greater than or equal to a threshold K . A subarray is a contiguous subsequence of days, and the arithmetic mean (arithmetic meme) is the sum of virality scores divided by the length of the subarray.

Input

The first line of input contains N ($1 \leq N \leq 10^6$) and K ($-10^9 \leq K \leq 10^9$), which are the number of days posted and the threshold respectively. The next line contains N integers $V_1, V_2, \dots, V_N$ ($-10^9 \leq V_i \leq 10^9$), which are the virality scores.

Output

Display the length of the longest viral streak. If there are no viral streaks, display 0.

Sample Input 1

```
5 3
4 5 6 7 8
```

Sample Output 1

```
5
```

Sample Input 2

```
6 4
1 2 10 8 1 2
```

Sample Output 2

```
6
```

Sample Input 3

```
4 1
-5 -4 -3 -2
```

Sample Output 3

```
0
```



Sample Input 4

7 5
9 1 1 1 1 10 10

Sample Output 4

4

Problem C

Corporate Family

Time limit: 8 seconds

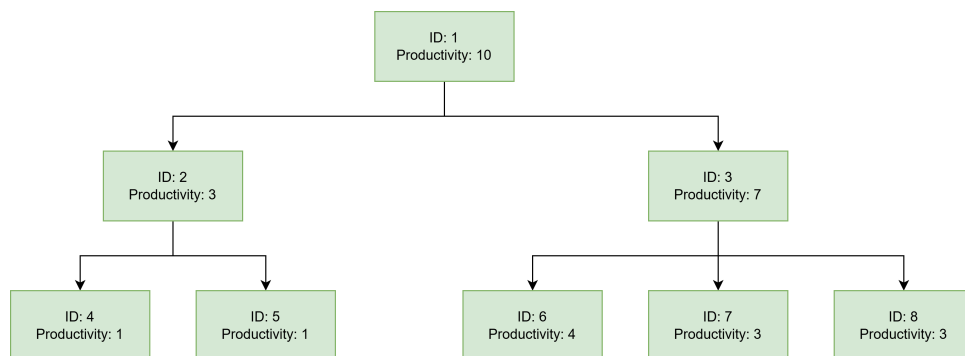
At Synergy Inc., we truly believe that every individual is a vital part of our organisational family. We are dedicated to nurturing positive relationships between employees and their managers and celebrating the unique contributions of each team member.

Every employee at our company is assigned a unique ID from 1 to N , with employee 1 serving as our inspiring CEO. Every other employee reports directly to exactly one manager, with that manager having a smaller ID number. Each employee also has a productivity value, a positive integer that reflects their work output.

Our company's *synergy* is a reflection of this collaborative spirit. It is defined as the number of pairs of employees who both share the same manager and have productivities that sum to the productivity of their manager.

For example, the synergy of the company shown in the diagram would be 3, because there are 3 pairs of employees whose productivities add to the productivity of their managers:

- Employee 2 + Employee 3 = Employee 1 ($3 + 7 = 10$)
- Employee 6 + Employee 7 = Employee 3 ($4 + 3 = 7$)
- Employee 6 + Employee 8 = Employee 3 ($4 + 3 = 7$)



As our employees grow and develop, their productivity may increase over time, which may change our company's synergy. After each productivity update, please report the current synergy of the company, so that we can continue to shine together as a team.

Wait... you're asking what we actually do here? Erm... let me align with the key stakeholders and circle back to you on that one.

Input

The first line of input contains a single integer N ($2 \leq N \leq 200\,000$), which is the number of employees in the company.

The next N lines describe the employees in order of their ID, from 1 to N . The i th of these lines contains two integers M ($1 \leq M < i$ or $M = -1$) and P ($1 \leq P \leq 10^8$), indicating that employee i reports to manager M and has a productivity of P . Only employee 1, the CEO, has $M = -1$, since they have no manager.

The following line contains a single integer I ($1 \leq I \leq 200\,000$), which is the number of productivity increase events.

The next I lines describe the increase in employee productivity. Each of these lines contains two integers E ($1 \leq E \leq N$) and X ($1 \leq X \leq 10^8$), indicating that the productivity of employee E increases by X . This increase is permanent: it adds to the employee's productivity from this point onward and is included in all subsequent synergy calculations.

Let W be the maximum number of employees under any manager. It is guaranteed that $W \times I$ is no greater than 200 000.

Output

For each productivity increase, display the company's synergy.

Sample Input 1

```
8
-1 10
1 3
1 7
2 1
2 1
3 4
3 3
3 3
3
4 1
2 5
1 5
```

Sample Output 1

```
4
2
3
```

Sample Input 2

```
6
-1 1
1 1
1 1
1 1
1 1
1 1
3
6 1
1 1
5 1
```

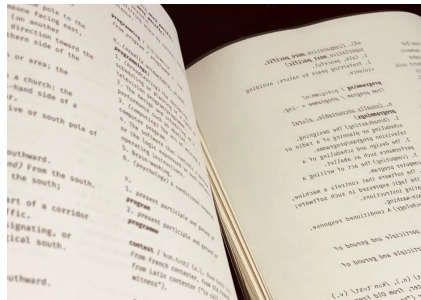
Sample Output 2

```
0
6
3
```

Problem D

Dictionary Shuffle

Time limit: 8 seconds



Source: Hadyn Tang - using work of Wiktionary authors, CC BY-SA 4.0

You have a list of N words, which will be used to create a dictionary in lexicographic (alphabetical) order. However, you are allowed to reverse some (possibly all or none) of the words before sorting. This means there are many possible dictionaries, depending on which words you choose to reverse.

For each word in the list, determine the earliest and latest index (starting from 0) at which the word could appear in a dictionary, either in its original form or reversed, across all possible dictionaries.

Input

The first line contains a single integer N ($1 \leq N \leq 300\,000$), which is the number of words in the list.

Following are N lines, each containing a single string S_i which is the i th word in the list. Each string has at least one character and consists of only lowercase letters.

The sum of the lengths of all strings does not exceed 1 500 000. No two words are the same, and no word is the reverse of another word in the list. A word may, however, be the same as its own reverse.

The list is not necessarily in lexicographic order.

Output

Display N lines. The i th line should contain two integers, denoting the earliest and latest index at which the word S_i or its reverse could appear in a dictionary.

Sample Input 1

```
3
apple
banana
racecar
```

Sample Output 1

```
0 1
0 1
2 2
```

Sample Input 2

```
3
a
ab
abc
```

Sample Output 2

```
0 0
1 2
1 2
```

Sample Input 3

```
6
cat
dog
level
noon
stats
top
```

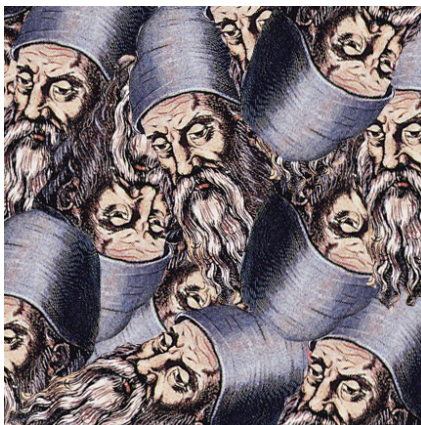
Sample Output 3

```
0 5
0 1
1 2
2 3
3 5
3 5
```


Problem E

Euclidean Extension

Time limit: 2 seconds



Source: Euclid, coloured woodcut, 1584, Public Domain (adapted)

A prime number has no integer factors greater than 1 other than itself. Euclid's proof that there are infinitely many primes relies on the fact that, given any set of primes, we can construct a number equal to 1 plus their product; this number has a prime factor not in the original set. For example, for the set $\{2, 17\}$, the number constructed is: $1 + (2 \times 17) = 35$, with prime factors 5 and 7.

Euclid's construction is a specific case of a more general form where a set S of primes is partitioned into two disjoint subsets ($S = S_1 \cup S_2$). Then, we compute the sum of the products of the two subsets ($\prod S_1 + \prod S_2$). Euclid's construction is the degenerate case where the pair of subsets consists of the empty set and the set itself. Note that the product of the empty set is 1 ($\prod \{\} = 1$) and the product of the set containing only the element x is x ($\prod \{x\} = x$).

For a given set S of primes, the numbers resulting from the possible partitions will be distinct. Define the resulting set as $E(S)$: the "extended Euclidean numbers" of S . Each number in $E(S)$ has the property that all of its prime factors are not in S .

Given a set S of primes, you are interested in generating new prime factors similar to Euclid's proof. Specifically, you want to find the set of all prime factors of numbers in the extended Euclidean numbers $E(S)$ of S .

For example, given the set $S = \{2, 3, 5\}$, its set of extended Euclidean numbers is $E(S) = \{11, 13, 17, 31\}$, where $11 = 5 + 2 \times 3$, $13 = 3 + 2 \times 5$, $17 = 2 + 3 \times 5$, and $31 = 1 + 2 \times 3 \times 5$.

In the previous example all the elements of $E(S)$ were prime, but this is not true in general. Say $S = \{2, 17\}$. Then $E(S) = \{35, 19\}$, where $35 = 1 + 2 \times 17$ and $19 = 2 + 17$. The resulting set of prime factors is $\{5, 7, 19\}$ (prime factors of 35 are 5 and 7, and 19 is prime).

Input

The input is a single line of integers representing the original set of primes. This line starts with N ($0 \leq N \leq 6$), which is the number of elements in the set. N distinct integers follow, each is a prime in the inclusive range from 2 to 127, and they are given in ascending order.

Output

Display a single line starting with an integer M , the number of distinct generated prime factors, followed by M integers describing the prime factors in ascending order.

Sample Input 1

0

Sample Output 1

1 2



Sample Input 2

2 2 17

Sample Output 2

3 5 7 19

Sample Input 3

3 2 3 5

Sample Output 3

4 11 13 17 31

Problem F

Firefighting Robots

Time limit: 5 seconds



Source: Milremrobotics, CC BY-SA 4.0

Firefreetown consists of just one long road with houses located at certain positions along it. Unfortunately, the town has no firefighters. The town wants to install special *firefighting robots* just in case a (very ironic) fire breaks out. Robots can only be installed at the same locations as houses because they need to be sheltered from the elements. The robots move at a speed of one metre per second.

Each house has a fire safety rating, indicating how many seconds it would take to burn down if it catches fire. A robot must be able to reach the house within this time to save it. A house with a robot installed in it is always safe.

What is the fewest robots the town needs to make sure no house will burn down if it catches fire? The town is only considering the case where a single house catches fire in their scenario.

Input

The first line of input contains the number of houses n ($1 \leq n \leq 300\,000$). The next n lines describe the houses. Each line contains two integers x_i ($0 \leq x_i \leq 10^9$) and s_i ($0 \leq s_i \leq 10^9$), the location of the i th house along the road (in metres from the start of the road) and the number of seconds it would take to burn down, respectively.

No two houses occupy the same location.

Output

Display one integer, the fewest robots the town needs.

Sample Input 1

```
3
0 2
2 1
4 2
```

Sample Output 1

```
1
```

Sample Input 2

```
2
5 0
1 4
```

Sample Output 2

```
1
```

Sample Input 3

```
4
1 10
3 1
5 2
10 5
```

Sample Output 3

```
2
```

Sample Input 4

```
2
1 8
10 8
```

Sample Output 4

```
2
```

Problem G

Get Their Attention

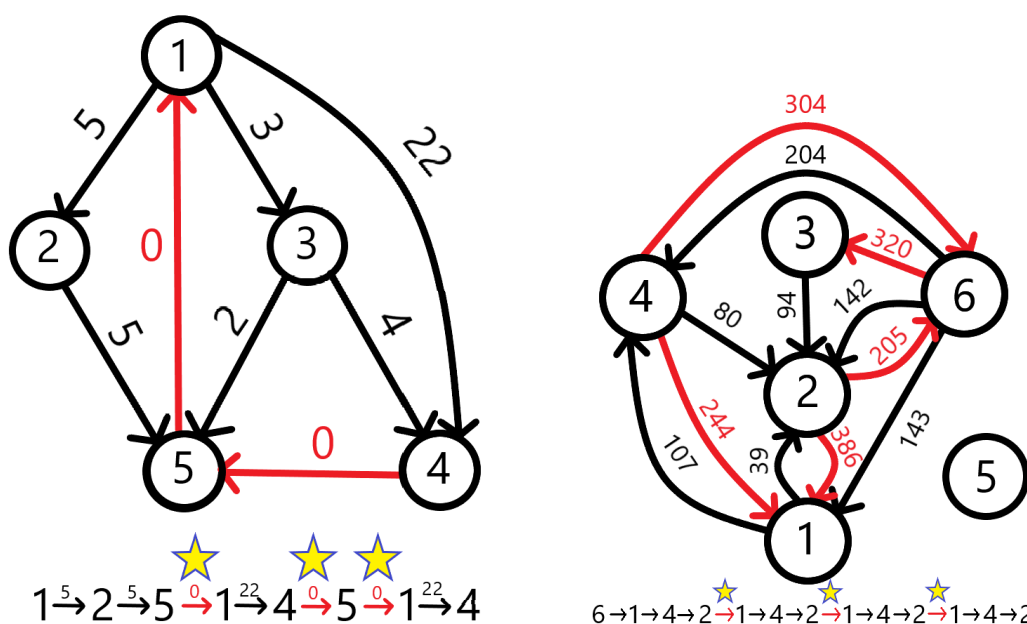
Time limit: 10 seconds

You are playing the hit game GTA (Get Their Attention). In this game, contrary to the title, you are driving around the city trying to *avoid* the attention of the police.

The map of the city is a collection of n locations, connected by m one-way roads. Each road has a certain number of points associated with it. If you drive down a road, you collect that many points. You can drive down individual roads multiple times to earn these points multiple times.

There are two types of roads in the city: quiet roads and busy roads. You are allowed to drive down both types of roads; however, driving down a busy road will incur one “star” rating with the police. Driving down a quiet road incurs no stars.

You would like to reach at least a particular score S by starting and ending at any node in the city, driving along a sequence of roads (possibly repeating some roads), and in the process incurring the *least* “star” rating with the police. Compute the smallest “star” rating possible to achieve at least a score of S .



The images above represent the maps for sample cases 1 and 2, as well as a solution path. Black edges are quiet roads, while red edges are busy roads.

Input

The first line of input contains three integers n ($1 \leq n \leq 50$), which is the number of locations, m ($0 \leq m \leq n^2 - n$), which is the number of roads, and S ($0 \leq S \leq 10^8$), which is the target score.

The next m lines describe the roads. Each of these lines contains four integers a ($1 \leq a \leq n$), b ($1 \leq b \leq n$, $a \neq b$), p ($0 \leq p \leq 10^8$), and r ($0 \leq r \leq 1$), indicating that there is a road from a to b , and that you will receive p points for driving on this road. If $r = 0$, then the road is a quiet road, otherwise the road is a busy road. No two lines will contain the same ordered pair a, b .

Output

If it is impossible to achieve at least S points, then display -1 . Otherwise, display the smallest “star” rating possible to achieve a score of at least S points.

Sample Input 1

```
5 8 54
1 2 5 0
1 3 3 0
1 4 22 0
2 5 5 0
3 5 2 0
3 4 4 0
4 5 0 1
5 1 0 1
```

Sample Output 1

```
3
```

Sample Input 2

```
6 12 2046
6 1 143 0
2 6 205 1
4 2 80 0
1 2 39 0
6 4 204 0
6 3 320 1
3 2 94 0
6 2 142 0
2 1 386 1
1 4 107 0
4 6 304 1
4 1 244 1
```

Sample Output 2

```
3
```

Sample Input 3

```
4 4 1000
1 2 0 0
2 3 0 0
3 1 0 0
2 4 999 0
```

Sample Output 3

```
-1
```

Sample Input 4

```
1 0 0
```

Sample Output 4

```
0
```

Problem H

Humanoid Robot Tango

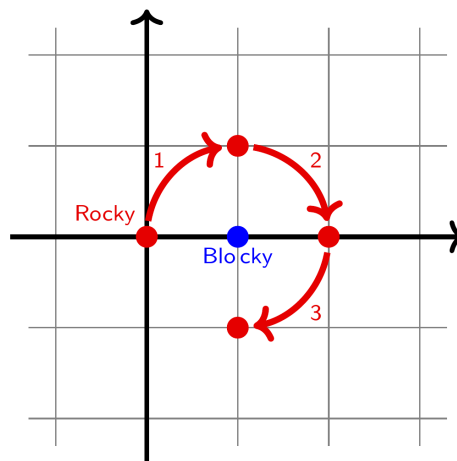
Time limit: 1 second

Rocky and Blocky are starting to choreograph a new dance—a Samba of Phlogiston Plus Cryogenics—which takes place on an infinite flat floor.

At the start of the dance, Rocky is at $(0, 0)$, and Blocky is at $(1, 0)$. Throughout the dance, one dancer is always designated as the active dancer and the other as the inactive dancer. Rocky is initially the active dancer.

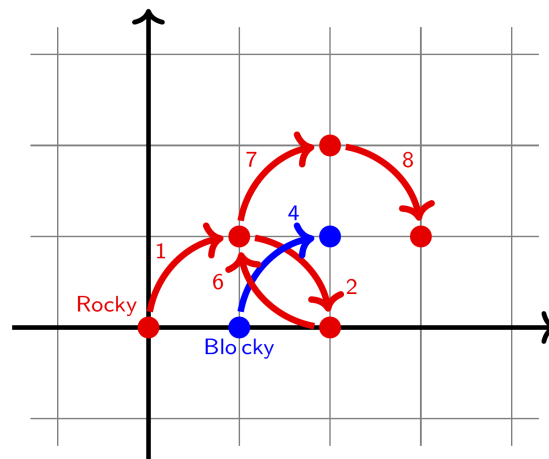
The dance consists of two different moves: slides (denoted by S) and bumps (denoted by B).

- In a slide, the active dancer rotates 90° clockwise around the inactive dancer at a fixed radius of 1 unit, while the inactive dancer remains in place. For example, if three consecutive slides occurred (which we would denote by SSS) Rocky's position would change as shown:



- In a bump, Rocky and Blocky swap their roles as the active and inactive dancer. That is, the person currently designated as the active dancer becomes the inactive dancer and vice-versa. They remain in the same positions during the bump.

By combining these types of moves, the dance becomes more complicated. The sequence of moves “slide, slide, bump, slide, bump, slide, slide, bump” (denoted SSBSBSSSB) would result in the following movements of the positions of the dancers:



Rocky is curious: at the end of the dance, where will he end up?

Input

The first line of input contains a single integer L ($1 \leq L \leq 10^6$), which is the number of moves in the dance. The second line contains a string with L characters which are either S or B, where the i th character is S if the i th move of the dance is a slide, or B if it is a bump.

Output

Display two integers, x and y , where (x, y) is Rocky's final position at the end of the dance.

Sample Input 1

```
3
SSS
```

Sample Output 1

```
1 -1
```

Sample Input 2

```
9
SSBSBSSSB
```

Sample Output 2

```
3 1
```


Problem I

I Get Extra Votes

Time limit: 4 seconds

The election is coming up. N candidates (numbered $1, 2, \dots, N$ by their pre-election rankings) are running to become the Supreme President and Really Cool Dude (SPRCD).



Source: Kurtkaiser, CC0 1.0 (adapted)

Each ballot has three different candidates listed, in order of preference. After all ballots are submitted (including any extra ones you might sneak in), the winner is determined by repeating the following process $N - 1$ times:

- Each ballot gives one vote to the highest-ranked candidate on it who has not yet been eliminated. If all candidates on the ballot have been eliminated, that ballot gives no votes.
- The candidate with the fewest votes is eliminated. If there are multiple candidates with the fewest votes, the one of those with the largest pre-election ranking is eliminated.

After this process, only one candidate remains. They are declared the winner (and immediately assume the office of SPRCD).

However, you are trying to destabilize the government by manipulating the voting system. Ballots are submitted electronically, and you have found a bug that allows you to submit exactly K identical copies of your ballot. You already know how everyone else is going to vote.

Determine all of the candidates that you can make win the election.

Input

The first line of input contains three integers N ($3 \leq N \leq 30$), which is the number of candidates, M ($1 \leq M \leq 30$), which is the number of other voters, and K ($1 \leq K \leq 10$), which is the number of identical ballots you submit.

The next M lines describe the other voters' ballots. Each of these lines contains three distinct integers v_1, v_2, v_3 ($1 \leq v_1, v_2, v_3 \leq N$), denoting that their ballot states that they prefer the candidate with pre-election ranking v_1 , then v_2 , and then v_3 .

Output

Display the pre-election rankings of all candidates that can win the election after you cast your K ballots. Display the numbers in ascending order.

Sample Input 1

```
3 3 1
1 2 3
1 2 3
3 2 1
```

Sample Output 1

```
1
```

Sample Input 2

```
3 3 1
1 2 3
3 2 1
3 2 1
```

Sample Output 2

```
1 3
```

Sample Input 3

```
4 3 1
3 4 1
2 3 4
2 4 3
```

Sample Output 3

```
1 2
```

Sample Input 4

```
4 3 10
4 2 1
4 2 1
4 2 1
```

Sample Output 4

```
1 2 3 4
```

Sample Input 5

```
9 4 2
3 1 4
1 5 9
2 6 5
3 5 8
```

Sample Output 5

```
1 2 3 5 6 9
```

Problem J

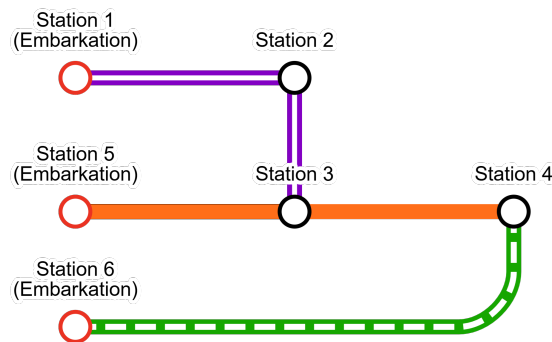
Jumping Between Stations

Time limit: 6 seconds

Metros are underground rapid-transit trains found in many big cities. They're fast, sleek, and hailed as the future of public transport. But no matter how cool metros may be, they cannot eliminate the frustration of transferring between lines. Moving from one train to another can be a hassle, and missing your stop can turn a short trip into a veritable pilgrimage!

Your local metro consists of one or more lines, and each of these metro lines visits one or more stations. Metro lines can be travelled in either direction. The first station on every metro line is unique. It is called that line's *embarkation station*. The embarkation station for each metro line is not on any other metro line (non-embarkation stations may or may not be on multiple metro lines). It is always possible to travel from every station to every other station using the metro lines.

A *transfer* occurs when a passenger switches from one metro line to another at a common station. Consider the following example metro map:



To travel from station 1 to station 6, you need to make at least two transfers. You could:

- Start at station 1 and take the purple (top) line to station 3.
- Transfer to the orange (middle) line and travel from station 3 to station 4.
- Transfer to the green (bottom) line and travel from station 4 to station 6.

Every station has an *accessibility rating*, R , which means that no matter which station you want to go to, you can get there from this station using at most R transfers. For example, station 5 has an accessibility rating of 1, since it can reach every station using at most one transfer.

A station is considered a *very accessible* station if no other station has a lower accessibility rating. In the example above, three stations (5, 3, and 4) are very accessible, each with an accessibility rating of 1.

Given a metro map, determine how many stations are very accessible and their accessibility rating.

Input

The first line contains two integers, n ($1 \leq n \leq 50\,000$) and m ($1 \leq m \leq 20$), where n is the total number of stations in the metro system, and m is the number of metro lines.

The next m lines describe the metro lines. Each of these lines starts with an integer k ($1 \leq k \leq n$), which is the number of stations on this line, then follows k distinct integers $x_1, \dots, x_k$ ($1 \leq x_i \leq n$), which are the IDs of the stations served by this metro line. x_1 is this metro line's embarkation station.

Each station is on at least one metro line.

Output

Display the number of stations that are very accessible and their accessibility rating.

Sample Input 1

```
6 3
3 1 2 3
3 5 3 4
2 6 4
```

Sample Output 1

```
3 1
```

Sample Input 2

```
5 2
3 2 1 3
3 4 5 3
```

Sample Output 2

```
1 0
```

Sample Input 3

```
9 5
2 1 2
3 3 2 4
3 6 2 5
3 8 7 4
3 9 7 5
```

Sample Output 3

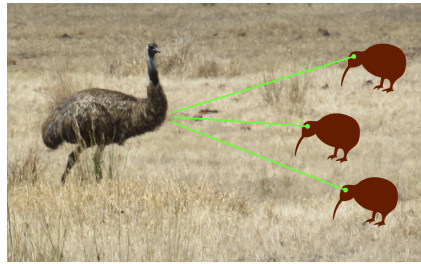
```
3 1
```

Problem K

Kiwi Invasion

Time limit: 5 seconds

You have recently become the Secretary of War of Emutopia. The President of your country wants to invade the neighbouring state of Kiwiland due to its exploitative trade policy. It is your job to advise the President on the likelihood that an invasion will succeed.



Sources: Calistemon, CC BY-SA 4.0; flomar, Public Domain (adapted)

Due to difficult economic conditions in both nations, neither Emutopia nor Kiwiland can afford any actual military equipment, so they will instead roll dice to determine the outcome of a conflict. Emutopia and Kiwiland have D_e and D_k dice respectively in their war chests. An invasion would take place in several rounds. If Emutopia and Kiwiland have t_e and t_k troops respectively at the start of a round, then:

1. Emutopia rolls $n_e = \min(D_e, t_e)$ dice and Kiwiland rolls $n_k = \min(D_k, t_k)$ dice. All dice are fair, S -sided dice, with sides numbered from 1 to S .
2. The values from the Emutopian dice are sorted in descending order, giving a sequence $E_1 \geq E_2 \geq \dots \geq E_{n_e}$. Likewise, the values from the Kiwilander dice are sorted in descending order, giving a sequence $K_1 \geq K_2 \geq \dots \geq K_{n_k}$.
3. The first $n = \min(n_e, n_k)$ of the two sequences are compared. For every $i \leq n$ such that $E_i > K_i$, one Kiwilander troop is sent home, and for every $i \leq n$ where $E_i \leq K_i$ one Emutopian troop is sent home. Thus, at the end of the round, Emutopia will have $t_e - |\{i = 1, 2, \dots, n : E_i \leq K_i\}|$ troops remaining and Kiwiland will have $t_k - |\{i = 1, 2, \dots, n : E_i > K_i\}|$ troops remaining.
4. If all of Kiwiland's troops have been sent home (i.e. $t_k = 0$) at the end of the round, the invasion succeeds. If, on the other hand, all of Emutopia's troops have been sent home (i.e. $t_e = 0$), the invasion is defeated. Note that it is not possible that both t_k and t_e reach 0 in the same round, or for either to become negative.

These rounds continue until the invasion either succeeds or is defeated.

For example, suppose $t_e = 4$, $t_k = 5$, $D_e = 3$, $D_k = 2$ and $S = 6$ meaning that Emutopia starts with 4 troops while Kiwiland starts with 5. In the first round, Emutopia rolls $n_e = \min(3, 4) = 3$ dice while Kiwiland rolls $n_k = \min(2, 5) = 2$ dice. Imagine Emutopia rolls 1, 5, 4 while Kiwiland rolls 2, 5, giving $E = [5, 4, 1]$ and $K = [5, 2]$. The top two rolls from each are now compared ($n = \min(3, 2)$). Since $E_1 \leq K_1$ ($5 \leq 5$) and $E_2 > K_2$ ($4 > 2$), both countries lose 1 troop. Now $t_e = 3$ and $t_k = 4$ with the invasion continuing for further rounds.

Given the number of Emutopian and Kiwilander troops at the beginning of the invasion, the number of sides on the dice and the number of dice each side gets, calculate the probability of the Emutopian invasion succeeding.

Input

The input consists of a single line containing five integers t_e ($1 \leq t_e \leq 1000$), which is the number of troops Emutopia starts with, t_k ($1 \leq t_k \leq 1000$), which is the number of troops Kiwiland starts with, D_e ($1 \leq D_e \leq 10$), which is the maximum number of dice Emutopia can roll, D_k ($1 \leq D_k \leq 10$), which is the maximum number of dice Kiwiland can roll, and S ($2 \leq S \leq 25$), which is the number of sides on each die.

Output

Display the probability of the Emutopian invasion succeeding.

Your answer should have an absolute or relative error of less than 10^{-9} .

Sample Input 1

1 1 2 2 2

Sample Output 1

0.250000000000

Sample Input 2

4 5 3 2 6

Sample Output 2

0.358606233202